Chapter 6, Problem 35P

Problem

An inductor has a linear change in current from 100 mA to 200 mA in 2 ms and induces a voltage of 160 mV. Calculate the value of the inductor.

Step-by-step solution

Step 1 of 2

The inductor voltage, $v = 160\ \text{mV}$

The inductor current changes from 100mA to 200 mA in 2 ms.

Determine the change in inductor current, $\frac{di}{dt}$.

$$\begin{aligned}\frac{di}{dt} &= \frac{200\ \text{mA} - 100\ \text{mA}}{2\ \text{ms}} \\ &= \frac{100\ \text{A}}{2\ \text{s}} \\ &= 50\ \text{A/s}\end{aligned}$$

Step 2 of 2

Write the formula for voltage across the inductor.

$$v = L\frac{di}{dt}$$

Substitute $160\ \text{mV}$ for v and $50\ \text{A/s}$ for $\frac{di}{dt}$.

$$160\times10^{-3} = L(50)$$

$$\begin{aligned}L &= \frac{160\times10^{-3}}{50} \\ &= 3.2\times10^{-3}\ \text{H} \\ &= 3.2\ \text{mH}\end{aligned}$$

Thus, the value of inductor is $\boxed{3.2\ \text{mH}}$.